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Recognition of PRI modulation using an optimized convolutional neural network with a gray wolf optimization based on internet protocol and optimal extreme learning machine

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In the modern electronic warfare (EW) landscape, timely and accurate detection of threat radars is a critical and necessary issue in electronic support Measure (ESM) and electronic intelligence (ELINT) because these radars' correct and timely detection plays an essential role in electronic countermeasures strategies. The PRI (pulse reputation interval) modulation type is one of the main parameters in radar signal analysis and identification. However, recognizing PRI modulation is challenging in a natural environment due to destructive factors, including missed pulses, spurious pulses, and large outliers, which lead to noisy sequences of PRI variation patterns. This paper presents a new four-step real-time approach to recognize six common PRI modulation types in noisy and complex environments. In the first step, an optimal convolutional neural network (CNN) structure was formed by a gray wolf optimization (GWO) based on the Internet Protocol (IP-GWO) according to the simulated PRI data set, which acts as a feature extractor. In the second step, the last fully connected layers are replaced by an extreme learning machine (ELM) to improve the time complexity of the proposed model. After that, in the third stage, GWO was introduced to adjust the biases and weights of ELM to reduce the complexity of the suggested method space. In the fourth step, the efficacy of the suggested approach is compared with the results of 5 CNNs based on transfer learning. The suggested strategy performs superior to other benchmarks, with 98.23% final accuracy on the simulated PRI dataset and 99.20% on the real PRI dataset. Also, the training time of the proposed method for 42,000 images takes 69.45 s, which indicates that the proposed method is real-time. The results show the optimal performance of the suggested approach for both simulated and real data sets. Compared to traditional DCNN models, which require significantly more computation and exhibit reduced performance under noisy conditions, the proposed method demonstrates enhanced resilience and efficiency for both simulated and real data sets. Therefore, the proposed method can be a suitable option in systems where accuracy and time are critical for detecting threat radars.

Keywords Pulse repetition interval modulation, Convolutional neural network, Extreme learning machine, Gray wolf optimizer

In the modern EW environment, timely and accurate detection of threat radars is one of the primary tasks of ESM and ELINT systems to determine the counter strategy. These systems are responsible for receiving and analyzing radar signals and, finally, radar navigation. PRI modifications constitute a fundamental element of radar signal analysis and a necessary source for their identification by these systems. Additionally, radar warning receivers, decoys, and jammers use PRI properties; therefore, detecting PRI types in radar signals is essential to identify threats and determine the strategy to deal with them.

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Nowadays, radars use different PRI modulations instead of a simple fixed pattern. Therefore, recognizing PRI modulation helps a lot to identify radars. However, recognizing PRI modulation is difficult due to destructive factors—missed pulses, spurious pulses, and large outliers—contributing to PRI modulation sequence noise. In addition, the variety of parameters within each category of PRI modulation has made the identification significantly more challenging. It is a complicated task to identify types of PRI modulation.

This research introduces a new method to recognize PRI modulation type, which aims to reduce temporal complexity and spatial complexity with a view to high recognition accuracy. For this purpose, the data sets used in all stages of training and evaluation of the proposed method are simulated with the inspiration of the real environment data and all the destructive factors affecting the recognition of PRI modulation have been considered. Finally, the simulation results show that the proposed method performs better than the existing methods. This can be seen in the Discussion section.

Comparison with related work

Extensive research has been conducted to solve the problem of PRI modulation recognition and classification. The existing methods can be generally categorized into three divisions: based on statistics, based on decision trees, and based on learning. The learning technique is divided into two: based on learning and deep learning[1,2]. In methods based on classical statistics, the histogram technique[3–6] is used to distinguish types of PRI modulation. Because these methods are simple, they can detect a small number of PRI modulations, and their performance is significantly reduced under noisy conditions[1,2]. Decision tree-based methods[7–14] require manual thresholds, which, in addition to being time-consuming, are very sensitive to the amount of noise and changes in PRI parameters in learning-based methods.[1,2] also require a detailed feature design and feature extraction process before using the neural network. These drawbacks prevent the methods from quickly adapting to changes in PRI modulation, and their disadvantages are pretty similar because they have a large amount of work in data preprocessing, and the results cannot adapt to the environment with a high ratio of missed pulses and spurious pulses[1,2].

Recently, deep learning[15–17] has emerged as a powerful tool for many classification tasks, and some researchers have used this technique for radar signal identification[16–21] because deep learning can automatically extract signal features. It has also had great success in many fields, such as image processing, speech recognition, object detection, and so on[22]. The concept of deep neural network (DNN) or deep learning was first proposed in reference[23] and then quickly became one of the most widely used topics in artificial intelligence in recent years. After that, Deep Belief Network (DBN)[24] and DCNN[25] algorithms were successfully proposed in various applications.

Recently, researchers have used DCNN to detect PRI modulation[26]. DCNNs are DNNs usually used to recognize and classify images and speech[27–29]. Because DCNNs can automatically learn and extract features from raw data, they have been effective in several applications[27–29].

DCNNs use the error backpropagation method to adjust the weights and biases, so their training takes time, especially when the number of data is large. To solve this problem, this paper uses a DCNN structure optimization. For this purpose, first, an optimal DCNN structure is formed by adjusting the hyperparameters by IP-GWO. Then the fully connected layers are replaced by an ELM. Next, in order to maintain the stability and strength of the network, the weights and biases of ELM are adjusted by GWO. In addition to increasing accuracy, this combination significantly reduces training time.

This distinction clearly highlights the efficiency and real-time capability of the proposed Roker compared to existing methods.

Research background

As stated, recently, researchers have used DCNN to detect PRI modulation[26]. DCNNs are DNNs usually used to recognize and classify images and speech[27–29]. Because DCNNs can automatically learn and extract features from raw data, they have been effective in several applications[27–32].

Researchers[33] have introduced a DCNN-based approach to identify seven types of PRI modulation. The outcomes of the simulation show that this approach has reached an overall accuracy of 96.1% in 50% of missed pulses and 20% of spurious pulses. The dataset used in this research was 25,000 samples.

In[1], the authors have used DCNN to categorize seven types of PRI modulation. The simulation results show that while the amount of missed pulses and spurious pulses is between 25 and 30%, the overall detection accuracy of the method is equal to 96%. The dataset in this research is 3,000 samples for each type of PRI modulation.

Reference[34] presents a new model that uses the intrinsic property of a temporal convolutional network (TCN). The simulation results show that this approach can precisely classify seven distinct types of PRI modulation; in this approach, the detection accuracy is more than 98% in the percentage of missed pulses and 30% unwanted pulses and large outliers (which are caused by scanning radar antennas are created) 0–10% has been achieved. The outcomes are obtained from 40,000 images, randomly selected from seven unique modulations, each with the same probability of selection.

Researchers used DCNN to classify eight modulation types in the study[26]. The simulation results indicate that the suggested method has reached an overall accuracy of 98.5%. In this method, the percentage of incorrect and unwanted pulses is 15%. The dataset contains 16,000 examples for each category of PRI modulation.

Introduces[2] a distinctive methodology grounded upon DCNN. This approach uses 60,000 accurate data to evaluate the proposed method. This method has achieved 97.05% accuracy in detecting five types of PRI modulation of accurate radar signals.

Previous studies demonstrated the potential of DCNNs to detect PRI modulation. However, these approaches have certain limitations—time-consuming training, mainly when the amount of input data is significant—which should be considered. Optimization methods are used in various cases to improve the performance of neural

networks. This article also uses this technique to improve the performance of the proposed method[2,35–40]. Therefore, this manuscript proposes a methodology that covers these limitations. The findings demonstrate the efficacy and resilience of the suggested approach and achieve high accuracy in the simulation and real datasets. This approach is carried out through the following process:

Main actions:

1. In the first step, IP-GWO constructs the optimal DCNN structure, which acts as an automatic feature extractor.
2. In the second step, the last fully connected layers are replaced by an ELM to improve the suggested model's time complexity.
3. In the third stage, GWO was introduced to adjust the biases and weights of ELM to reduce the complexity of the proposed model space.
4. The suggested method attains an enhanced ultimate accuracy of 98.23% on the simulated dataset and 99.20% on a real dataset of six common PRI modulations. It needs only 69.45 s to execute 42,000 training images to demonstrate its real-time abilities.

Background knowledge

This section briefly describes PRI, DCNN, GWO, and ELM.

Pulse repetition interval

A radar system transmits a "pulse" signal to a target. The PRI is the interval between one pulse and the start of the next. Figure 1 shows the transmitted and received pulses on radar. The inverse of PRI is called pulse repetition frequency (PRF), expressed in frequency units (Hz). Each series of pulses, known as a pulse train, is a continuous time sequence of data. The PRI pattern is commonly referred to as PRI modulation. Various PRI modulations exist, which can be grouped into distinct groups. PRI is an essential parameter because the target and radar performance are revealed in many cases[2,39,41] by identifying its modulation.

The subsequent equation shows the generic function for a PRI cycle containing M pulses. Here, $\boldsymbol{t_n}$ is the TOA of the pulses. A transmitter specifies how long to wait before transmission of a new pulse and thus uses several implementations of the Function $\boldsymbol{F}(\boldsymbol{n})$[2,41]:

$$F(n) = t_n + 1 - t_n, n \in Z_n \tag{1}$$

Radars generally use six main types of PRI modulation[34,39], as described in Table 1.

Deep convolutional neural network

A DCNN is a feed-forward NN that combines convolutional, pooling, and fully connected layers. This type of network exploits a notable spatial correlation by using a local connection pattern between the neurons of adjacent layers[39,43]. The architecture of the DCNN is illustrated in Fig. 2.

1. First, an image is received by the input layer.

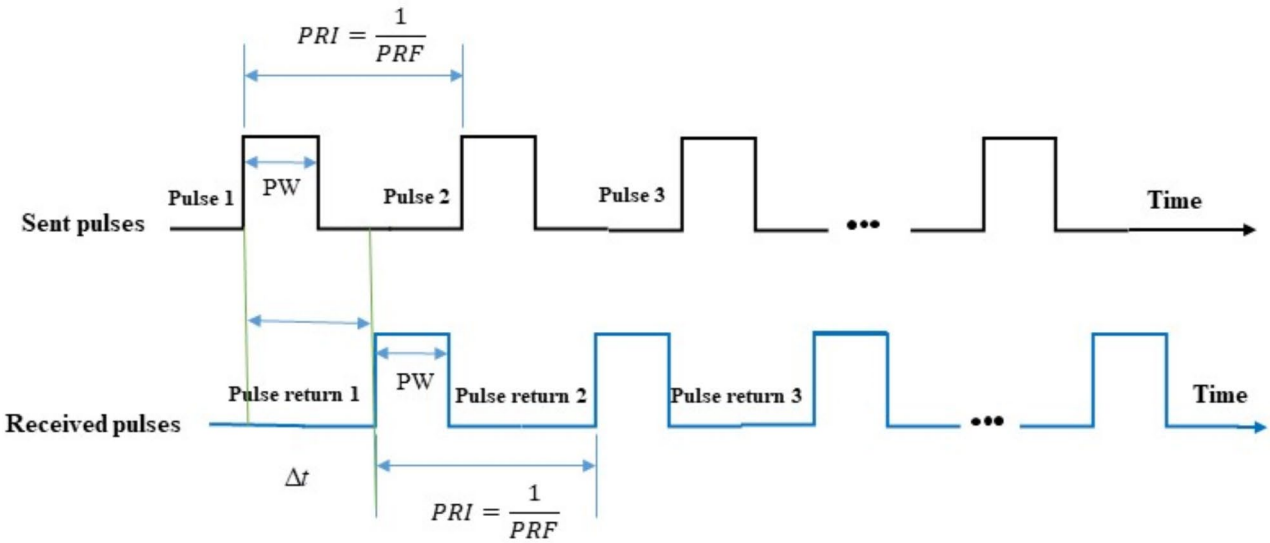

Fig. 1. Transmitted and received pulse train in a radar[42].

R	PRI modulation type	Specifications
1	Constant	Changes are usually seen in fixed-type PRIs. The average value of fixed PRIs is more than 1% higher than the typical value of variable PRIs. This type of PRI has high stability, which corresponds to pulse Doppler systems (MTI).
2	Stagger	In stagger-type PRI, several fixed PRIs are repeated on a pulse-to-pulse basis and in a periodic pattern, eliminating the effects of blind speed in MTI systems and increasing target detection accuracy.
3	Dwell and Switch (D&S)	In D&S type PRI, clusters of pulses, each with a fixed PRI that changes from one cluster to another, are used to remove ambiguity in calculating the speed or range of targets and increase radar detection accuracy.
4	Periodic	In Periodic PRI, sinusoidal variations of up to 5% of the average PRI with frequencies up to 50 Hz or more are used for missile guidance, jamming, and ranging in military radars.
5	Jitter	Jitter-type PRI can vary considerably, up to about 30% of the average PRI, which is used to resist anti-radar targets (radar jamming) and reduce interference with other signals.
6	Sliding	In sliding type PRI, the max PRI (at minimum height) is typically less than six times the minimum PRI. (At max height) used to provide constant height coverage while reviewing the elevation angle or avoiding jamming.

Table 1. Types of PRI modulation[34,39].

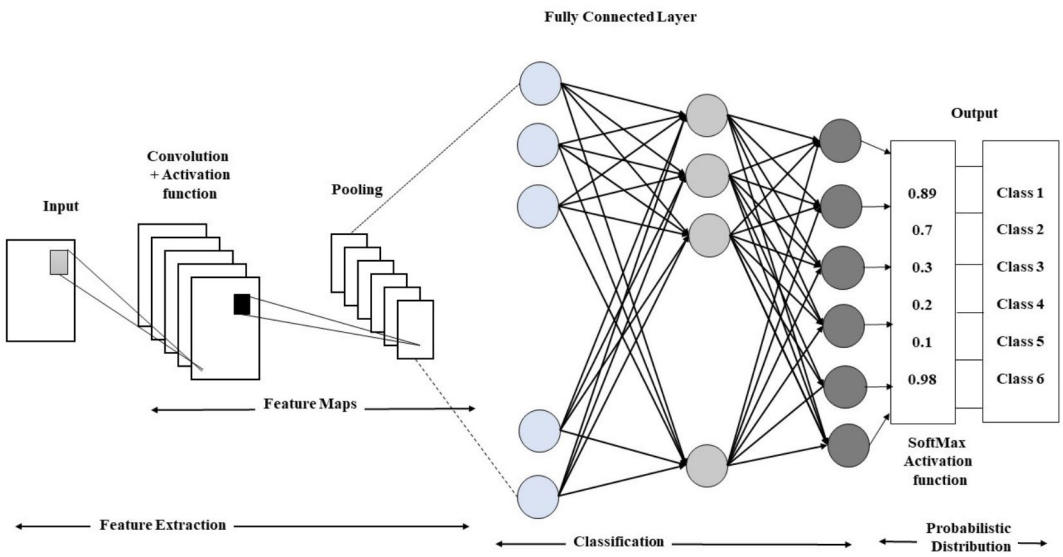

Fig. 2. DCNN architecture[44].

2. The convolutional layer performs a convolution operation on the input and then transmits the resulting output to the subsequent layer. This technique facilitates the network's ability to concentrate on specific local locations and acquire diverse properties.
3. The activation layer is typically implemented after each convolutional layer. The model incorporates a non-linear activation function, such as a rectified linear unit (ReLU), which enables it to acquire knowledge from the error and adapt accordingly.
4. The pooling layer is positioned after the activation layer and conducts a down-sampling operation across the spatial dimensions. This process effectively reduces the input's dimensionality, decreasing the computing complexity. Max Pooling and average pooling are widely used pooling layers in various neural network architectures.
5. The wholly connected layer consists of neurons that establish connections with all activations in the preceding layer, similar to conventional neural networks. On a more general level, they can be conceptualized as classifiers.
6. The output layer produces the final result of the neural network.

One of the most successful DL models for image analysis is DCNN.

Extreme learning machine

The perceptron neural network has two major drawbacks in training time: The first drawback is that it uses gradient descent to adjust the weights, making the training process time-consuming, especially when the training data is extensive! The second drawback of this network is the large number of parameters[45]. In this network, many parameters must be set in the training process, which makes the training time very high. The ELM neural network has provided a straightforward approach to solving this problem and has a very high learning speed. This network has the same structure as RBF but generally adjusts one parameter during training. Unlike RBF, where the synaptic weights between the input layer and the hidden layer are fixed, and the value is one, the input layer is connected to the hidden layer with a series of weights in this network. Of course, the

advantage of ELM is that in this network, the weights are assigned a random value. They are assigned first and do not need to be adjusted during training. The hidden layer's neurons are expected, and there is no need to find the centers and sigma of each neuron. Finally, the only adjustable parameter of this network is the synaptic weights between the hidden layer and the output layer. It is a forward network and calculates the synaptic weights in one moment using the pseudo-inverse method. And this has made the learning speed of this network very high. The noteworthy point here is that the performance of this algorithm is very high. Although the number of adjustable parameters is less, outstanding performance has been reported in the articles for this algorithm[46,47]. Figure 3 shows the structure of ELM[46,47].

As shown in Fig. 3, ELM consists of an input, hidden, and output layer.

The formula of the extreme learning machine

The activation function can be expressed as follows[46,49]:

$$\boldsymbol{T}_j = \sum_{i=1}^{l} O_i f\left(w_i, bi, x_i\right) \tag{2}$$

In this context, the symbol w_i indicates the weight of the input connection, b_i represents the bias of the ith hidden neuron, x_i represents the weight of the output connection, and $\boldsymbol{T}_j$ signifies the final output of the ELM. Equation (1) can be expressed in matrix form, as seen by Eq. (3)[46,49]:

$$\boldsymbol{T}^T = \boldsymbol{HO} \tag{3}$$

$\boldsymbol{T}^T$ denotes the transposition of matrix **T**. The matrices **H** and **O** can be expressed as follows[46,49]:

$$\boldsymbol{H} = \begin{bmatrix} f\left(w_1, b_1, x_1\right) f\left(w_2, b_2, x_1\right) & \cdots & f\left(w_l, b_k, x_1\right) \\ \vdots & \ddots & \vdots \\ f\left(w_1, b_1, x_\beta\right) f\left(w_2, b_2, x_\beta\right) & \cdots & f\left(w_l, b_k, x_\beta\right) \end{bmatrix} \tag{4}$$

$$\boldsymbol{O} = \left[O_1, O_2, \ldots, O_k\right]^T \tag{5}$$

The principal aim of ELM training is to minimize training errors. The classical ELM method uses input biases and weights that are determined randomly and employs an infinitely differentiable activation function. This is done by selecting the output weight, ***O***, using the least-squares value in Eq. (5). The corresponding solution can be seen in Eq. (6) above when this is the case, as demonstrated in References[46,50].

$$\underset{O}{min} \left\| \boldsymbol{H}O - \boldsymbol{T}^T \right\| \tag{6}$$

$$\widehat{O} = \boldsymbol{H}^{+}\boldsymbol{T}^T \tag{7}$$

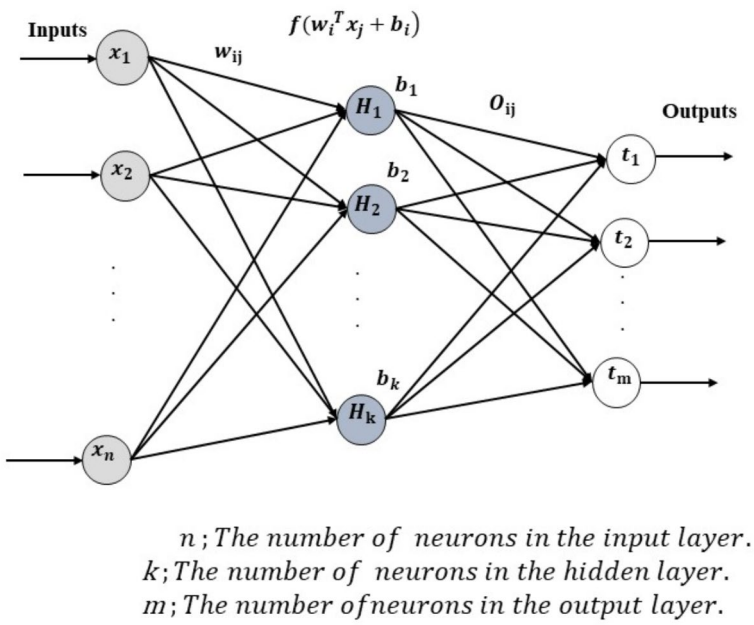

Fig. 3. The structure of ELM[47,48].

The matrix $\mathbf{H}^{+}$ denotes the revised Moore–Penrose inverse of matrix $\mathbf{H}$.

Gray wolf optimization

GWO is one such metaheuristic algorithm based on grey wolf hunting behavior. This population-based algorithm is easy to work with and can be tailored to fit large-scale problems easily. In this algorithm's hierarchy of needs, grey wolves are apex predators. This algorithm can solve issues like setting hyperparameters and other advanced optimization tasks. In many instances, it aids in delivering optimal solutions with optimal speed and accuracy[50,51].

The formula of the gray wolf optimization

The optimization process demonstrates the importance of selecting the right solutions. Adding a restriction of a maximum of three solutions to the problem allows us to resolve even the most complex issue. This method helps us avoid going down the wrong track and instead find the best solution to the problem. It also enables us to investigate a broader range of possibilities and guarantee that the solution chosen is optimal and the best for that specific issue[50,51].

The gray wolf behavior algorithm is an inspiration for studying social behavior. Such wolves, in their social hierarchy, treat each other with respect. The social hierarchy of these wolves is determined based on the ability and strength of each individual. These behaviors lead to an effective hunting mechanism because each wolf in the group, with high cooperation and coordination, can achieve more extensive hunts. In other words, cooperation between wolves is one of the main factors of hunting success and survival for the gray wolf. A gray wolf is a social being that lives and shares its survival activities with others in a well-structured society. Among these social units, an alpha wolf (α) is selected to be the leader and has the highest social status within the group. This wolf defends the group from intruders' attacks and has the privilege of accessing significant food materials. Other pack members, including beta (β), delta (δ), and omega (ω) wolves, determine their place in the hierarchy based on their strength and social ranking. Alphas are of the highest order and act as group leaders. They are responsible for guiding and maintaining order in the herd. The beta wolf seems to be the one who helps the alpha wolf in pack discipline and even supports his decisions. However, ranking even below them are respected Delta Wolves who, though they seem to have power from within, lack leadership qualities. The Last, on top of the pecking order, the lowest, the omega wolf, possesses no influence and often ends up taking care of the younger wolves. Such a pecking order and division of roles is an excellent aid to communication and interrelations within the herd, which is important for the herd in terms of survival and growth. Gray wolves are among the animals that possess and utilize the most sophisticated and specialized hunting scheme. It is interesting to note that the members of the wolf community possess extreme[50,51].

These animals hunt in groups, coordinate, and use clever methods to separate the prey from the herd. First, wolves collectively surround the prey, and then one or two wolves chase and attack the prey while the rest of the pack pursues the killer. Intelligence and are highly coordinated, as exemplified by their successful hunting techniques. This together sheds light on the capabilities of the people of this region.

Most logical connections follow the following formulas[50,51]:

$$\overrightarrow{D} = \left|\overrightarrow{C} \cdot \overrightarrow{P}_P(t) - \overrightarrow{P}(t)\right| \tag{8}$$

$$\overrightarrow{P(t+1)} = \overrightarrow{P}_P(t) - \overrightarrow{A}.\overrightarrow{D} \tag{9}$$

t represents the run, A and C are the coefficient vectors, Pp represents the position vector of the prey, and P is the position vector of the wolf. A and C are equal to[50,51]:

$$\overrightarrow{A} = 2\overrightarrow{a}\,\overrightarrow{r}_1 - \overrightarrow{a} \tag{10}$$

$$\overrightarrow{C} = 2 \cdot \overrightarrow{r}_2 \tag{11}$$

$$\overrightarrow{a} = 2 - 2\frac{t}{Max_iter} \tag{12}$$

In Evolutionary Strategies, the value of the vectors is decreased linearly from 2 to 0 with an increase in attempts. The component is usually randomly generated, and for every wolf and effort, vector A helps balance the exploration and exploitation levels[50,51].

On the other hand, vector C always adds some degree of randomness. This is essential because the optimization may get stuck in the local optimum. To avoid this issue, we need meta-heuristics and random factors. These methods promote the possibility of achieving a more optimal solution and increase the efficiency and accuracy of algorithms[50,51].

The Pp vector is dependent on 3, the best solution, and the updating formulas of each of the agents (wolves) are as follows[50,51]:

$$\overrightarrow{D}_\alpha = \left|\overrightarrow{C}_1 \cdot \overrightarrow{P}_\alpha - \overrightarrow{P}\right|, \overrightarrow{D}_\beta = \left|\overrightarrow{C}_2 \cdot \overrightarrow{P}_\beta - \overrightarrow{P}\right|, \overrightarrow{D}_\delta = \left|\overrightarrow{C}_3 \cdot P_\delta - \overrightarrow{P}\right| \tag{13}$$

$$\overrightarrow{P}_1 = \overrightarrow{P}_\alpha - \overrightarrow{A}_1, \overrightarrow{P}_2 = \overrightarrow{P}_\beta - \overrightarrow{A}_2, \overrightarrow{P}_3 = \overrightarrow{P}_\delta - \overrightarrow{A}_3 \tag{14}$$

$$\vec{P}(t+1) = \frac{\vec{P}_1 + \vec{P}_2 + \vec{P}_3}{3} \tag{15}$$

The P vector is considered the current location of the agent. Then, the vector P(t + 1) is updated using the calculated formula. The present wolf adjustment forms part of the three most optimal wolves in the previous iteration; however, the three optimal wolves do not have to be precisely equal to vector P(t + 1)[50,51].

Because of the vector C, a small random displacement is added to the new position. This action avoids getting stuck in the local optimum, which can improve the optimization performance[50,51].

Finally, the GWO pseudo-code is as follows[50,51]:

Initialize the wolf population, $i = \overline{1, n}$
Initialize α, A, and C
Calculate the fitness of each search agent
  $P_\alpha = the\ best\ search\ agent$
  $P_\beta = the\ second\ best\ search\ agent$
  $P_\delta = the\ third\ best\ search\ agent$
While t< max number of iterations do
  for each search agent, do
    Randomly initialize $\vec{r}_1$ and $\vec{r}_2$
    Update the position of the current search agent by Equation (15)
  Update α, A, and C
  Calculate the fitness of all search agents
  Update P_α, P_β, and P_δ
  t=t+1
return P_α

Algorithm 1. GWO.

The algorithm's time complexity depends on the number of iterations, the available factors, and the size of the vectors. In this situation, however, they can usually be estimated as O (k, n) where k is the number of iterations and n is the number of agents.

Proposed method

In this method, first, the best DCNN structure is formed according to the dataset simulated by IP-GWO. Since DCNN networks in their last layer use a fully connected MLP, because the dimensions of the input data are high, as a result, the number of weights and biases will also be significant, and considering that the decreasing gradient is used to adjust the weights, adjusting the weights and the training process will take time. This study uses an ELM network instead of the final MLP network for acceleration. The ELM network randomly selects the weights of the input layer, and the ELM can generate these parameters before seeing the learning data. The results will be unstable since ELMs choose their weights and biases randomly. In this research, the weights and biases of the ELM network are determined by GWO optimizer algorithms to increase the stability and strength of the network. Therefore, in this article, GWO is used for two purposes: first, to determine the DCNN structure's hyperparameters and, finally, to adjust the weights and biases of ELM.

As a result, this paper introduces a new strategy that employs optimized DCNN as a feature extractor, ELM for rapid recognition of PRI modulation, and GWO method to enhance the system's reliability. The stated objective of this approach is to replace the weaknesses or the identified gaps of the existing methods concerning the effectiveness and reliability of detecting PRI modulation in the presence of destructive factors. Furthermore, developing more comprehensive data sets containing a wider range of PRI signals with different percentages of disruptive agents can help build and evaluate PRI modulation detection systems.

Two datasets—simulated and real data—are used to measure the suggested model's performance and compare it to the outcomes of five transfer learning-based CNNs.

IP-based gray wolf optimizer

Every device connecting to a computer network is assigned its identifying number, an internet protocol (IP) address[52]. A sub-network or subnet is used to determine the specific network an IP address belongs to, and it's done by "adding the starting address and the number of bits of the network mask in CIDR notation"[45]. The end IP address for a standard network range with a starting address and mask length of 192.168.1.0/8 is 192.168.1.255, in which the subnet range is 192.168.1.0 to 192.168.1.255. A network is identified through the GWO; the encoding method uses a binary string instead of a set of decimal numbers (with periods separating each group of three decimal numbers) as the norm. Integer values seated within a CNN layer correspond to constant values constituting extensive binary strings representing numerous shorter binary strings, each depicting a single feature.

The binary output from the GWO-implemented CNN layers is said to be valid since it meets the default requirements. It employs large numbers of chimpanzees as one of the vector parameters, which increases the complexity of the process of finding the best solution. This method divides a high numerical identifier into

Layer type	Layers parameter	The number of bits assigned to each parameter	Range of parameters
Convolutional	Filter size	3	[1,8]
	Stride size	2	[1,4]
	Feature maps	7	[1,128]
Pooling	Kernel size	2	[1,4]
	Type: (max pooling = 0), (average pooling = 1)	1	[1,2]
	Stride size	2	[1,4]
	Place holder	7	[1,128]
Fully-connected	No. neurons	11	[1,2048]

Table 2. The properties of the layers in the DCCN.

Layer type	Subnet (CIDR)	IP range
Convolution	54.0/4	54.0–69.255
Fully-connected	70.0/5	70.0–77.255
Pooling	78.0/5	78.0–85.255

Table 3. The four subnets have been employed for the four different levels of CNN.

Layer type	Binary	IP address	Layers parameter
Convolution	(0000)101,010,101,010	10.222	Filter size = 6
			Stride size = 3
			Feature maps = 56
Fully-connected	(00000)10,101,111,101	21.63	No. Neurons = 1405
Pooling	(00000)11,011,010,111	27.23	Kernel size = 3
			Type: Average Pooling = 1
			Stride size = 2

Table 4. The instances of IP addresses.

decimal values smaller than 256. Each of these values is then allocated a one-byte IP address representation rather than utilizing a giant integer to represent the ID of a device. By partitioning the binary phrase in this manner, the GWO vector is transformed into a vector consisting of individual byte components. This reduces the query size for each divided size as all parameters in each round of GWO updates are trained concurrently. In the forthcoming segment, we will carry forward these concepts to develop novel GWO encryptions. These encryptions would help to encode multiple levels into a GWO, thus simplifying the learning.

Gray wolf optimizer encoding strategy

The structure of the Network IP address affects the IP-GWO encoding methodology. The CNN design is composed of 3 levels, where each layer has a different set of parameters, leading to a considerably big number of parameters in every layer.

Table 2 provides data on the various categories of strata. An approach to consider is creating a network IP address with a predetermined length, which may then be divided into many CNN layers.

Before commencing any encoding process, it is imperative to ascertain the exact size of the binary string. The Parameter column in Table 2 displays the essential characteristics of the convolutional layer. Table 2 shows that a layer is characterized by twelve bits, while an IP address unit is characterized by either a byte or eight bits. The IP address's large size is that each of its 12 bits requires two bytes of storage.

As shown in Table 3, the subnets of all DCNN layers must be defined in CIDR style by their respective bit numbers. Due to the categorization of DCNN layers into three distinct categories, three subnets are necessary to handle all three coatings effectively. Table 3 displays the subnets and their respective configurations.

To accommodate the varying lengths of DCNN frameworks, particular levels in the IP-coded GWO vector are deactivated to modify the predetermined length of GWO after it has been initialized. These bits are selected from three distinct types of CNN layers, resulting in a final IP address range of 32.0/5, as presented in Table 3.

Table 4 demonstrates that the IP addresses in Table 4 consist of binary sequences from a particular layer. These strings are padded with zeroes till they achieve a two-byte length. Once the binary line runs this length, it is allocated a subnet mask and transformed into an IP address by splitting one byte with periods.

Figure 4 shows how an IP address is formed and vice versa.

Every layer must convert into a two-byte IP address format to ascertain the position of GWO. Hence, it is imperative to establish specific specifications, such as maximum length and maximum level of connectivity.

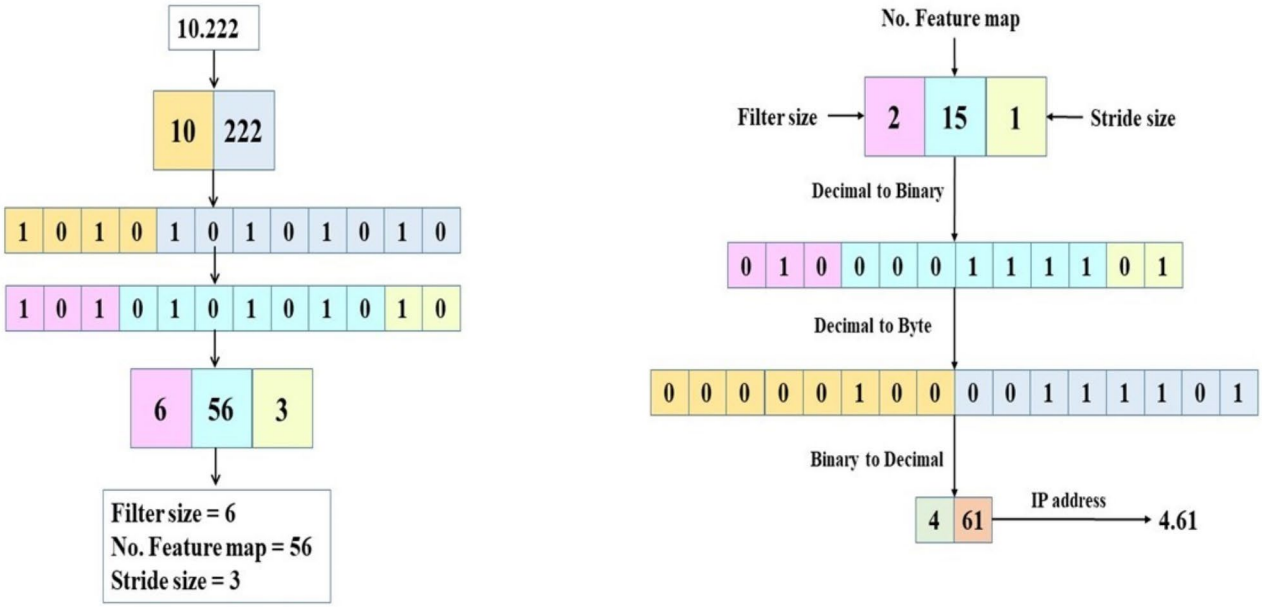

Fig. 4. The instances of IP addresses and vice versa.

Parameter	Max.value
DCNN layers	15
Size of population	25
Fully connected layers	3
Training epoch number	15
Batch size	100

Table 5. The comprehensive list of parameters.

Layer	Feature map	Size	Filter/Kernel size	Stride size	Activation function
Input		224 × 224			
Convolution	5	220 × 220	5 × 5	1	Relu
Convolution	33	73 × 73	4 × 4	3	Relu
Avg. pooling	33	24 × 24	2 × 2	3	–
Fully-connected	-	1513	-	-	Relu
Fully-connected	-	500	-	-	Softmax
Output		6	-	-	
Learning rate	1e-5				
Epoch number	5				
Searching agents number	20				
Bach size	16				

Table 6. The best DCNN structure is created according to the dataset simulated.

Table 5 provides a concise overview of these issues. The array carrying position data will have a total length double the size of an individual byte for every byte, indicating one aspect of the GWO.

Table 6 shows the best DCNN structure created according to the dataset simulated in this approach.

Data set

The approach outlined in the preceding section is trained and evaluated using a dataset that mimics the pertinent natural environment. To simulate the actual scenarios, the scanning effects of the radar antenna were regarded first. These events result in substantial changes in two consecutive TOAs (compared to the original PRI), possibly regarded as numerous thousands of adjoining MP. Although these outliers are low (approximately 1–10%), they

can significantly reduce the recognition accuracy[34]. In contrast to prior studies within this domain, we tackle the issue of PRI modulation recognition by considering all practical disruptive influences.

Another crucial concern is the quantity of pulses utilized in the classifier mechanism. When the modulation is intricate or the time is substantial, detecting the modulation type can only occur if adequate pulses are received. Based on intuition and actual evidence, a range of 100–250 pulses is deemed sufficient for this function[34].

The input PRI range is another crucial problem. In references[3,4,6,7,33,53], average PRIs were simulated to evaluate the proposed methods. However, a wide range of PRIs must be analyzed in a true ESM or ELINT system. Accurate PRI radars are often between 2 µs to 50 ms.

However, initially, it may not appear too severe since the input data is normalized during preprocessing[30]. Nevertheless, this matter becomes more apparent when the issues of missed pulses and spurious pulses happen. Generally, the lower the PRI amounts, the graver the consequences of unwanted pulses, and the higher the PRI amounts, the increasingly intense the repercussions of missed pulses[34].

Considering the previous debate, it is necessary to incorporate the impacts of omitted and undesired pulses into the PRI vectors for all types of modulations. An acceptable range for the negative consequences is 5–30%[34]. Nevertheless, the combined total of the two percentages must not exceed 50%, as it is required that at least 50% of the members of a PRI vector pertain to the current circumstance[34]. Table 7 shows the general characteristics of the proposed realistic scenarios in this paper.

Equation (1) states that the characteristics of each modulation type should be considered when generating experimental data.

PRI modulation has six distinct types, and the definitions that follow are drawn straight from theses[41,54]; researchers conducted their thesis on the classification of PRI sequences using PRI:

$$Sin\,PRI : F(n) = PRI_0 + Asin\left(f * x_n + \varphi\right), PRI_0 > 0 \tag{16}$$

A is the amplitude, f is the frequency, φ is the phase, and x_n is the value determining sampling accuracy according to n.

$$Fixed\,PRI : F(n) = k, k > 0 \tag{17}$$

Here, the PRI is often constant.

$$Jitter\ PRI : The\,PRI\,is\,random\,with\,a\,mean\,value,\ \mu,\ and\,some\,spread,\ \sigma. \tag{18}$$

The random variable PRI has a mean value of μ and a range of σ.

$$Sliding\,PRI : F(n) = PRI_0 + \delta * (n\,mod\,M), PRI_0 > 0, \delta \in R. \tag{19}$$

$PRI_0 > 0$ is the initial amount of the sliding period $\delta \in R$ represents the PRI change rate during the sliding period, and M is the number of pulses in each sliding window.

$$Stagger\,PRI : F(n) = y_i, y_i > 0, i = n\,mod\,M, \tag{20}$$

PRI is an iterating series with fixed M amounts.

Technical details	Value
Range of PRI	2(µs)–50 (ms)
Number of pulses	100 –250
Unintentional tremor %	0–1
Large outlier %	1–10
Missed pulses %	0–5
	5–10
	10–15
	15–20
	20–25
	25–30
Spurious pulses %	0–5
	5–10
	10–15
	15–20
	20–25
	25–30

Table 7. General characteristics of the proposed realistic scenarios.

D&S PRI consists of multiple amounts determined by the function F(x). Within every sequence, the PRI dwells on a value (Dwell) with the same value $y_i > 0$ and number $x_i - x_{i-1}$ before switching to the next value y_{i+1}.

$$F(x) = \begin{cases} y_1, 0 \leq (n \bmod x_M) \leq x_1, \\ y_2, x_1 \leq (n \bmod x_M) \leq x_2, \\ \vdots \\ y_M, x_{M-1} \leq (n \bmod x_M) \leq x_M. \end{cases} \tag{21}$$

Table 8 shows the simulated data set used in this research. It is worth mentioning that the percentage of significant outliers in all data is 1–10%.

Figure 5 shows a case of PRI modulation types without destructive factors.

As Fig. 5 shows, the radars alternate the PRI pulse with a pulse or a group of pulses. This solves blind speed and unambiguous range problems and adds anti-tear capability.

In the real environment, pulse sequences are not so clean, and the effects of destructive factors can be seen in them.

Test and results

To train the proposed model, Intel-corei7-13700K, 32GB RAM, and Nvidia RTX 3080 hardware are used to implement the network. The desired models have been implemented using TensorFlow and Cross Libraries in the Python programming language. Since training time is essential in this research, the number of epochs was considered the same as the number of epochs set by IP-GWO for training DCNNs, i.e., 5 epochs (according to Table 6). Since acceptable outcomes were not obtained compared to the proposed method, the number of epochs was increased to 50 until analogous results were obtained compared to the suggested method, and the batch size was 16 for all networks (according to Table 6). The dataset is partitioned into three subsets: 70% for training, 15% for validation, and 15% for testing. The weights of the pre-trained networks for transfer learning are quantified using the method provided by the TensorFlow library. The network's training rate value of 0.00001 is considered.

Evaluation measurements

Classification performance of our method is evaluated using five commonly used criteria[2,39,55]. The. Each of these criteria has a specific meaning. For this reason, in this research, we use all five criteria to assess the efficacy of the suggested approach. The formulas provided in Contact[56] can be utilized to calculate them:

$$Recall = \frac{TP}{TP + FN} \tag{22}$$

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \tag{23}$$

$$F_1 - score = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \tag{24}$$

$$Precision = \frac{TP}{TP + FP} \tag{25}$$

$$MCC = \frac{TP \times TN - TP \times FN}{\sqrt{(FP + TP) \times (FP + TN) \times (FN + TP) \times (FN + TN)}} \tag{26}$$

TN indicates the total count of genuine negative instances, while TP denotes the quantity of positive events. FP refers to the count of false-positive samples, whereas FN means the count of false-negative events.

The proposed method's performance is evaluated on two simulated and real PRI data sets by comparing it with the performance of five CNN networks based on transfer learning.

The number of resulting images	Spurious pulses %	Missed pulses%	Large outliers%
6,000 images (each type of modulation 1000 images)	0	0	0
6,000 images (each type of modulation 1000 images)	0–5	0–5	1–10
6,000 images (each type of modulation 1000 images)	5–10	5–10	1–10
6,000 images (each type of modulation 1000 images)	10–15	10–15	1–10
6,000 images (each type of modulation 1000 images)	15–20	15–20	1–10
6,000 images (each type of modulation 1000 images)	20–25	20–25	1–10
6,000 images (each type of modulation 1000 images)	25–30	25–30	1–10

Table 8. The simulated data set was utilized in this research.

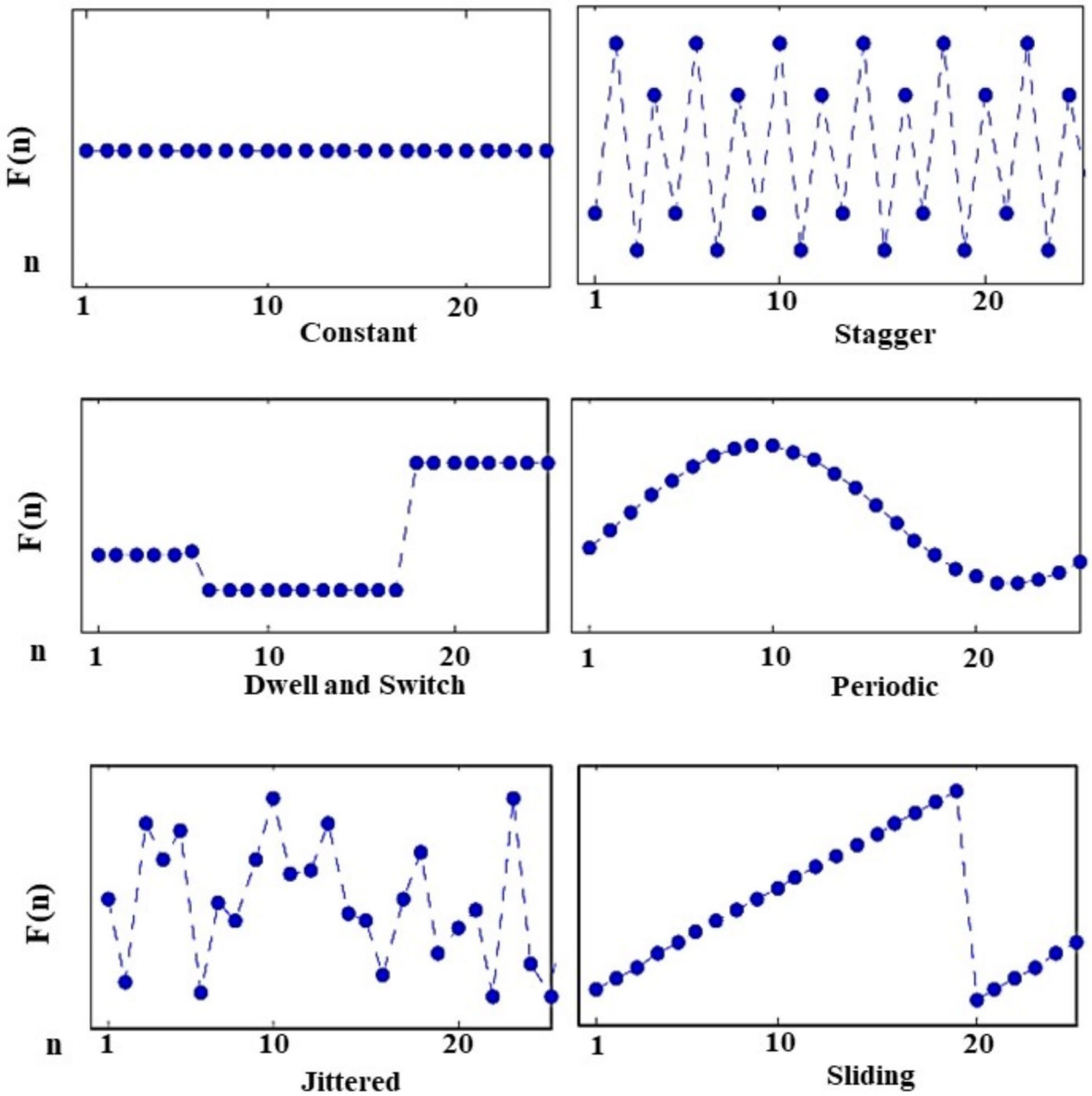

Fig. 5. Examples of PRI modulation types.

Analysis of recognition performance

Initially, to evaluate the proposed method, the data sets were selected so that the total number of destructive factors does not exceed 50% because at least 50% of the members of a PRI vector must correspond to correct pulses. Finally, an evaluation was done with different percentages of harmful factors separately.

Table 9 and Fig. 6 demonstrate the training time and accuracy of the suggested technique and five other CNNs over five epochs.

As it is clear from Table 9 and Fig. 6, in Epochs number 5, compared to the other five DCNN architectures, the proposed method has shown significantly better accuracy and training time in identifying PRI modulation. The accuracy of the proposed method for actual and simulated data is 99.20% and 98.23%, respectively, which is higher than all 5 DCNN models and shows the high ability of this model to identify all types of PRI modulations. While other models require significant training time, the proposed method has significantly optimized the calculations with a training time of 69.45 s.

As Table 9 and Fig. 6 clearly show, the DCNNs' results differ significantly from those of the suggested technique at the five epoch level. Therefore, in the continuation of the research, the number of epochs was increased to 50 so that the DCNNs' results are comparable to those of the suggested method.

Epochs number = 5				
			Accuracy (%)	
R	CNN	Training time (s)	Simulated dataset	Real dataset
1	VGG16	732.97	91.38	94.81
2	ResNet50V2	742.36	92.37	96.27
3	InceptionV3	746.12	91.11	95.14
4	Xception	733.73	90.66	95.38
5	MobileNetV2	741.68	90.92	95.24
6	**Proposed method**	**69.45**	**98.23**	**99.20**

Table 9. Comparison of the accuracy and training time of the suggested technique and different DCNN networks to identify PRI-modulation in the number of 5 Epochs.

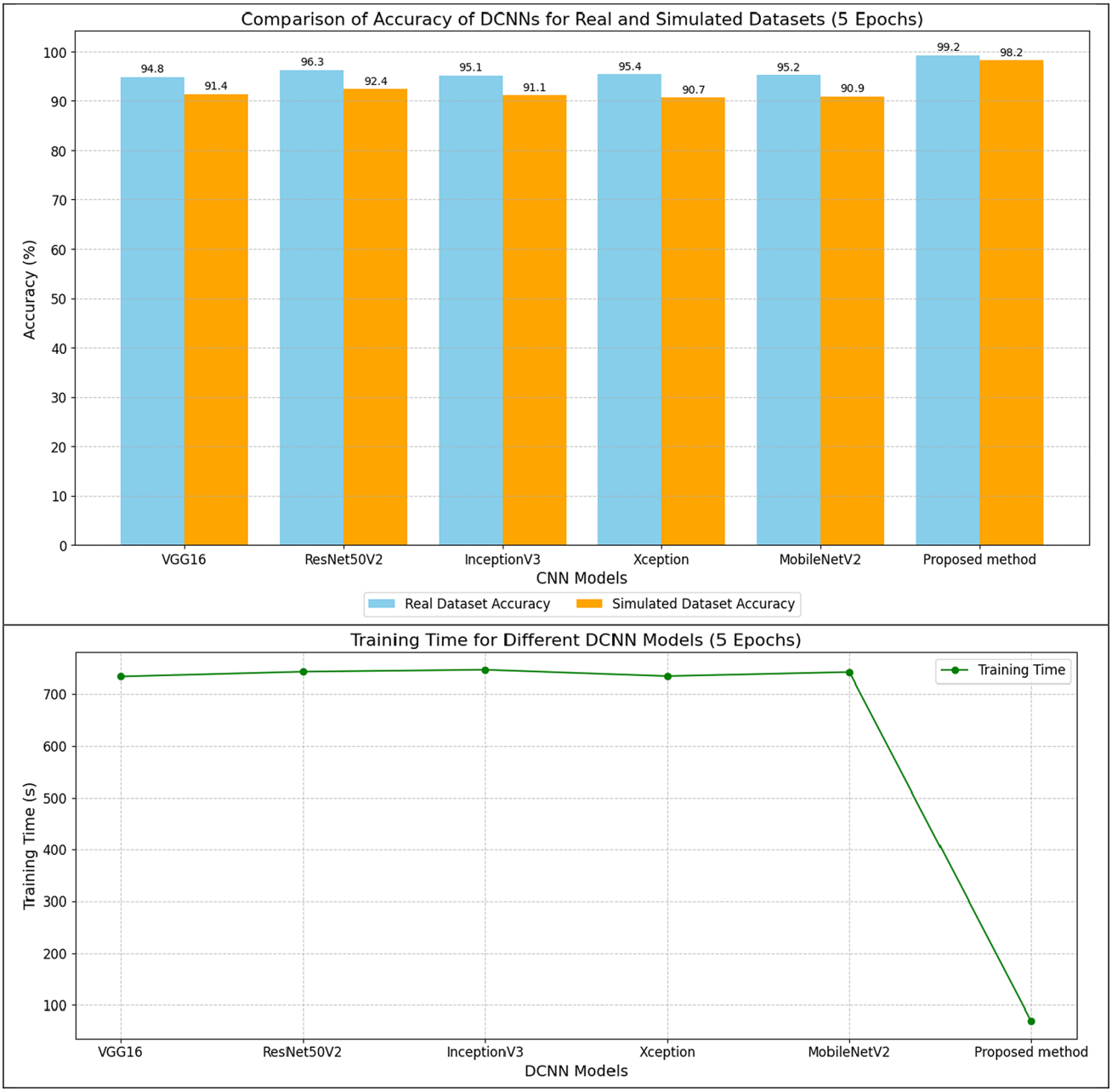

Fig. 6. Comparison of the accuracy and training time of the suggested technique and different DCNN networks to identify PRI-modulation in the number of 5 Epochs.

Epochs number = 50				
R	CNN	Training time (s)	Total parameters(million)	Trainable parameter (million)
1	VGG16	7335.15	57.59	12.87
2	ResNet50V2	7411.05	74.97	51.41
3	InceptionV3	7451.79	48.05	26.24
4	Xception	7336.61	72.27	51.41
5	MobileNetV2	7404.27	34.40	32.16
6	**Proposed method**	**69.45**	**0.7596**	**0.000003006**

Table 10. The complexity analysis of the proposed method and five other DCNN networks over 50 epochs on the simulated data set.

Epochs number = 50						
		Measurement criteria				
R	CNN	Accuracy (%)	Sensitivity (recall) (%)	Precision (%)	F1 score (%)	MCC (%)
1	VGG 16	95.19	95.23	95.32	95.22	94.32
2	ResNet50V2	95.23	95.28	95.33	95.28	94.37
3	InceptionV3	94.54	94.57	94.63	94.56	93.52
4	Xception	94.28	94.43	94.49	94.43	93.35
5	MobileNetV2	93.53	93.41	93.61	93.38	92.17
6	**Proposed model**	**98.23**	**98.09**	**98.28**	**98.18**	**97.79**

Table 11. The efficacy outcomes of the suggested technique and five other DCNNs (in 5O epochs) on the simulated dataset.

The following Tables 10, 11 and Figs. 7, 8 show the performance results of the proposed method and five other DCNN networks over 50 epochs on the simulated data set.

It is seen from Tables 10 and 11 and Figs. 7 and 8 that compared to DCNN models, the suggested technique has a much smaller number of trainable parameters (0.000003 million) and a very short training time (69.45 s) and performs better in identifying PRI modulations. While other models with trainable parameters and very high training times require heavier calculations, the proposed method shows its superiority by maintaining high accuracy in all evaluation criteria, including MCC, F1 score, Precision, Recall, and Accuracy. With an accuracy of 98.23% compared to other models (with accuracies between 93.53% and 95.23%), this model is considered the most efficient method, especially for real-time applications, and due to the lower need for computing resources, it is a suitable option in operational environments. is.

Next (Table 12 and Fig. 9), the efficacy outcomes of the suggested technique and five other DCNNs (in 50 epochs) are shown on the real dataset[2].

Table 12 and Fig. 9 show that the suggested model has performed much better and more efficiently than other CNN models in identifying PRI modulation on the real data set. With an accuracy of 99.20%, this model has the highest accuracy value and performs best in all evaluation criteria.

Compared to the other models, VGG16 and ResNet50V2 rank next with close accuracies (98.70% and 98.68%, respectively) but are still inferior to the proposed method's performance. In particular, the MobileNetV2 model, with an accuracy of 97.41%, is significantly far from the accuracy of other models, indicating that this model may have challenges in accurately identifying PRI modulations.

Table 13 and Fig. 10 show that the proposed method was evaluated separately on the data set with destructive factors.

The simulation results show that the accuracy of PRI modulation recognition of models decreases significantly with the increase in the percentage of destructive factors. However, the accuracy of the proposed method decreases at a much lower speed, which indicates the better performance of the proposed method against destructive factors.

Discussion

The evaluation of the proposed models shows the method's remarkable performance in identifying modulation PRI in different conditions. The use of IP-GWO in setting and optimizing DCNN hyperparameters and the use of WGO to set weights and biases of ELM have significantly increased the accuracy and efficiency of the model. Table 14 and Fig. 11 present the suggested model's performance compared with other techniques for comparison and evaluation. This research shows that the proposed method has provided good accuracy compared to other models and can help to increase the efficiency of radar and electronic warfare detection systems.

As it is clear from the Table 14 and Fig. 11, the suggested methods have not considered all the harmful factors, or they have considered a low percentage of them, or they have taken into account a certain percentage. Also, several articles in the table have not considered the percentages of destructive factors by the real systems; for example, in ESM and ELINT systems, to detect the type of PRI modulation, a maximum of 50% of the PRI

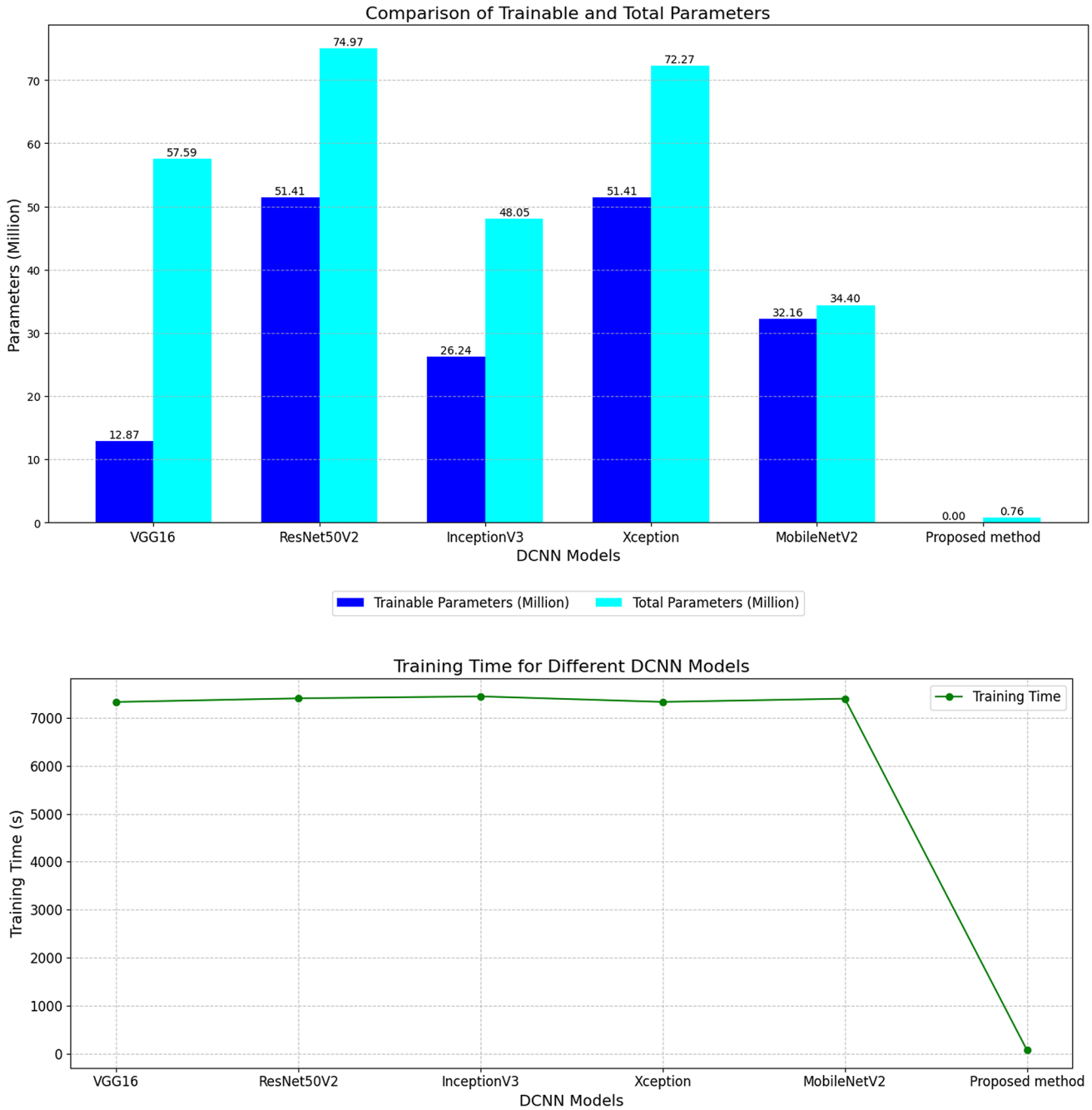

Fig. 7. The complexity analysis of the proposed method and five other DCNN networks over 50 epochs on the simulated data set.

sequence can be a set of destructive factors, which is not observed in some articles. In this article, destructive factors are considered according to real environment data.

As the table shows, the proposed method is accurate in detecting the type of modulation and performs better than the previous method in detecting real data. Therefore, the research shows that the proposed method provides good accuracy compared to other models and can help increase the efficiency of radar systems and electronic warfare detection.

Conclusions

This paper proposes a new four-step approach to recognize six common PRI modulation types. In the first step, the best DCNN structure is formed by a GWO based on IP-GWO according to the simulated PRI data set, which acts as a feature extractor. In the second phase, the last fully connected layers are substituted by an ELM to improve the time complexity of the suggested model. After that, in the third stage, GWO was introduced to adjust the biases and weights of ELM to reduce the complexity of the proposed model space. In the fourth step,

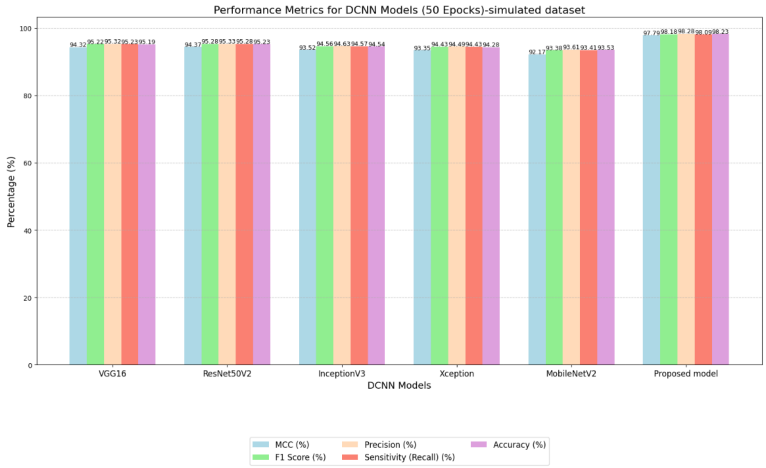

Fig. 8. The efficacy outcomes of the suggested technique and five other DCNNs (in 5O epochs) on the simulated dataset.

Epochs number = 50						
		Measurement criteria				
R	DCNN	Accuracy (%)	Sensitivity (recall) (%)	Precision (%)	F1 score (%)	MCC (%)
1	VGG 16	98.70	98.79	98.78	98.78	98.56
2	ResNet50V2	98.68	98.72	98.72	98.72	98.48
3	InceptionV3	98.52	98.46	98.45	98.45	98.16
4	Xception	98.30	98.36	98.36	98.36	98.05
5	MobileNetV2	97.41	95.50	95.43	95.44	94.54
6	**Proposed model**	**99.20**	**99.14**	**99.03**	**99.13**	**98.95**

Table 12. The efficacy outcomes of the suggested technique and five other DCNNs (in 50 epochs) on the real dataset.

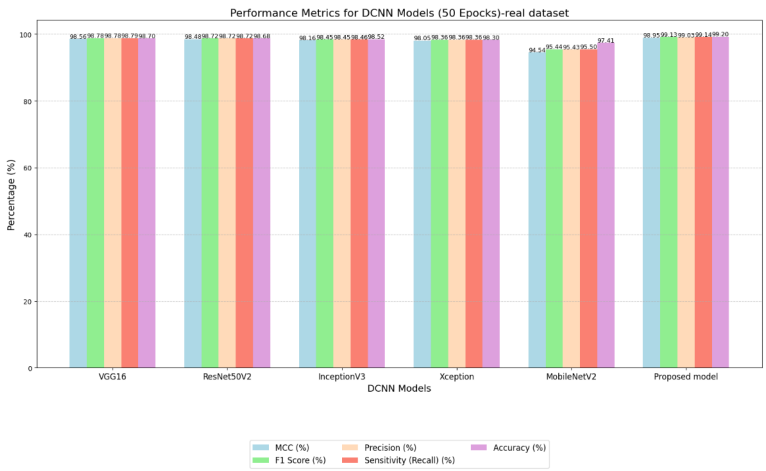

Fig. 9. The efficacy outcomes of the suggested technique and five other DCNNs (in 50 epochs) on the real dataset.

the efficacy of the suggested method is assessed on the simulated PRI data set. The efficacy of the suggested method was validated through a comparative analysis using five transfer learning-based DCNN networks on both simulated and real datasets. The suggested model performs better than other benchmarks, with 98.23% final accuracy on the simulated PRI dataset and 99.20% on the real PRI dataset. Also, the time required to train the proposed model for 42,000 training images is only 69.45 s. In the proposed method, due to the use

R	DCNN	Missed pulses (%) & Spurious pulses (%) Large outliers (1–10%)	Accuracy (%)
1	VGG16	0	99.98
		0–5	98.78
		5–10	97.38
		10–15	94.56
		15–20	93.16
		20–25	91.80
		25–30	91.31
2	ResNet50V2	0	99.96
		0–5	98.76
		5–10	97.40
		10–15	94.96
		15–20	93.20
		20–25	91.60
		25–30	91.36
3	InceptionV3	0	99.98
		0–5	98.60
		5–10	95.96
		10–15	94.29
		15–20	91.96
		20–25	91.09
		25–30	90.56
4	Xception	0	99.89
		0–5	98.38
		5–10	95.39
		10–15	94.31
		15–20	92.16
		20–25	91.62
		25–30	90.87
5	MobileNetV2	0	99.96
		0–5	97.49
		5–10	95.00
		10–15	93.53
		15–20	90.42
		20–25	89.42
		25–30	89.56
6	**Proposed method**	**0**	**100**
		0–5	**99.78**
		5–10	**99.50**
		10–15	**97.96**
		15–20	**97.20**
		20–25	**96.98**
		25–30	**96.67**

Table 13. Evaluation and comparison of the proposed method with the percentages of harmful factors separately.

of the combination of ELM and GWO, the temporal and spatial complexity is reduced compared to previous methods. The outcomes present the suggested model's high efficiency in processing simulated and real data and confirm its superiority in practical applications. Especially in fields where accuracy and sensitivity are critical in identification, the proposed method is considered a suitable option for use in complex electronic warfare systems by providing higher accuracy and better performance.

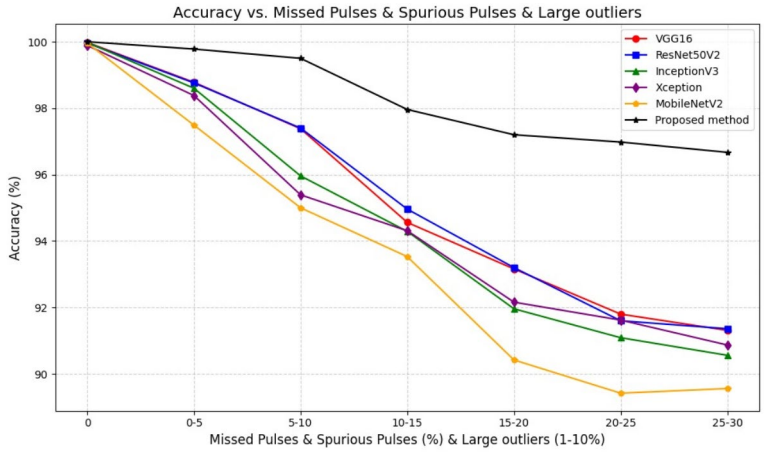

Fig. 10. Evaluation and comparison of the proposed method with the percentages of harmful factors separately.

R	References	The proposed method	Accuracy (%)	Destructive factors		
				Large outlier %	Spurious pulses %	Missed pulses %
1	[14]	NN with one hidden layer	99.95	---	---	10
2	[9]	Feed-forward NN with one hidden layer	99	--	25	15
3	[57]	Feed-forward NN	99.59	--	0–10	0–10
4	[58]	DMTL	72.3	--	30	30
5	[20]	RNN	99.91	--	--	2–30
6	[53]	ARNN	95.18	--	70	50
7	[59]	DCNN	96.1	--	20	50
8	[59]	DCNN	96	--	25–30	25–30
9	[34]	TCN	More than 98	1–10%	30%	30%
10	[26]	DCNN	98.5	--	15%	15%
11	[39]	Improved DCNN	98.80	Real data-PRI Sound		
12	[2]	Improved DCNN	97.05	Real data		
13	**Proposed method**	**Optimal DCNN**	**99.20**	**Real data [2]**		
			98.23	**1–10%**	**0–30%**	**0–30%**

Table 14. The proposed method's performance is compared with other techniques.

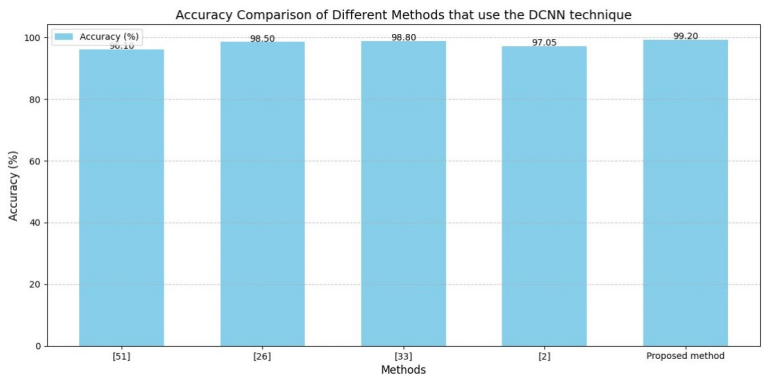

Fig. 11. Comparing the best performance of the proposed method with other DCNN-based techniques.

Data availability

The data supporting this study are openly available at the following: DOI: https://doi.org/10.17632/pj4h58n83j.1

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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